MIDTERM SYLLABUS

week	Theory	ZTB Youtube Videos	ZTB Slides	Mumit sir Slides
week 0	1. Intro to time complexity, RAM Model 2. Asymptotic Time Complexity	N/A	https://docs.google.com/presentation/d/1625N8yGQx9Gwn_AtcwwBuoB6Q7TvpArGJBuqTW0SZrE/edit?usp=drive_link	Assympt: https://drive.google.com/file/d/1rTgD7Vzsvd4jWFr9taPqrdttOHlHXRJY/view?usp=sharing
week 1	1. Time complexity (contd.) 2. Searching Linear / Binary / Ternary Search	N/A	https://docs.google.com/presentation/d/1TgsesHQHXpdNGv9xt3s3yJY3ULakcbGwzjRMPZuMFgs/edit?usp=drive_link	Iterative time complexity: https://drive.google.com/file/d/1rTgD7Vzsvd4jWFr9taPqrdttOHlHXRJY/view?usp=sharing Linear: https://drive.google.com/file/d/1vAJ2EbjdNmv6_ZgbEcDwDJj3RnwPs7qp/view?usp=sharing Binary: https://drive.google.com/file/d/1P6QjoYShah1YlFzdgNCRbnXd379BiuQb/view?usp=sharing Ternary: https://drive.google.com/file/d/1tP_ebyE3uHpGto6NIcDf5-VIzUsgoyOe/view?usp=sharing
week 2	1. Bubble, Insertion, Selection sort: best case, worst case analysis (self study, labs) 2. Merge Sort, Quick Sort, Optional: Count sort 3. Heap data structure and heap sort (labs)	N/A	https://docs.google.com/presentation/d/1CL2xRjy0CyAnTNK8abY_rmlRei6yi_1Os1MeKfEmQFs/edit?usp=drive_link	Merge: https://drive.google.com/file/d/1JUGTeiVhRnwaMKVcvwrukIKGymlq-U25/view?usp=sharing Quick: https://drive.google.com/file/d/1vbcbOtf5jkfIJBLzMbniFr3FZIvF3nbO/view?usp=sharing Heap sort: https://drive.google.com/file/d/1pE_xnkGCzSONOhGTs7kA5En5vgA_X2On/view?usp=sharing
week 3	1. Divide and conquer: Count inversion (from merge sort - lab), Karatsuba's multiplication algorithm, Max sum subarray, Optional: finding closest pair of points	N/A	Karatsuba: https://docs.google.com/presentation/d/1xcTFIl_wxMCVomD54YsKQxytc6VfNaf3zwEsliv7jWs/edit#slide=id.g251b994a15a_0_125 Max Sum Subarray: https://docs.google.com/presentation/d/1QKHH2QdwEnKPDgZmakmd-l3kaON3tfMjmzNaZDZ-Xbg/edit?usp=drive_link	Max subarray: https://docs.google.com/presentation/d/1rIC6OXh7W5ELsxkfGbnYWJRSv870Y3u2/edit?usp=sharing Karatsuba: https://docs.google.com/presentation/d/1xcTFIl_wxMCVomD54YsKQxytc6VfNaf3zwEsliv7jWs/edit?usp=sharing
week 4	1. Recursive time complexity, Master theorem 2. Graph basics (Adjacency List, Matrix, Space complexity and Time complexity, Symmetricity property, Dense graph, Sparse graph, Indegree, Outdegree)	Recursive time complexity and Master Theorem video: https://youtu.be/V9aRycrMaDY	Graph Basics: https://docs.google.com/presentation/d/1iiPEgybMYzdUxCEW1-Y4Bkd2j-cjYLhcuGNphnWoKfc/edit?usp=drive_link	Recursive Time complexity: https://drive.google.com/file/d/12cR_JBR8wZg0NDAll-wzL6fUsZLYcfQM/view?usp=sharing Graph intro: https://drive.google.com/file/d/1OkLO91GV4M23e-1EtvFr9yeGWgk6V91q/view?usp=sharing
week 5	1. BFS and applications: shortest path 2. DFS, Edge classification 3. Cycle detection, bipartite/bicolorable graph	BFS, Cycle Detection video: https://youtu.be/U_shdDPFw-k?feature=shared Bipartite/Bicolorable Graph video: https://youtu.be/8kz1eJg5S9w	BFS: https://docs.google.com/presentation/d/1LELld0WvSQlj4RwHIRbXshbUxO9RTY4m80Kq2MBbtt4/edit?usp=drive_link DFS: https://docs.google.com/presentation/d/1HkiUQheCF_NY7Sw3Af5X45lGb9rOyWFwWoHokfa6nE0/edit?usp=drive_link	Bfs Dfs: https://drive.google.com/file/d/1OkLO91GV4M23e-1EtvFr9yeGWgk6V91q/view?usp=sharing

FINAL SYLLABUS

week	Theory	ZTB Youtube Videos	ZTB Slides	Mumit sir Slides
week 7	1. DAG, Topological sort, Strongly Connected Components (Kosaraju)	DFS, DAG, Topological Sort, SCC (Kosaraju): https://youtu.be/oezlsdlGTUY	DFS, DAG, Toological Sort, SCC (Kosaraju): https://docs.google.com/presentation/d/1HkiUQheCF_NY7Sw3Af5X45lGb9rOyWFwWoHokfa6nE0/edit?usp=drive_link	bfs dfs: https://drive.google.com/file/d/1OkLO91GV4M23e-1EtvFr9yeGWgk6V91q/view?usp=sharing top sort: https://drive.google.com/file/d/1U-PVynHjD-PytCeHvcqtCsCaEE-iL-qc/view?usp=sharing SCC: https://drive.google.com/file/d/1fckjavRyW4yCAi2zmAjsF3pKhIsr3FJb/view?usp=sharing
week 8	1. Shortest paths: Dijkstra, 2. Negative cycle: Bellman-Ford 3. MST: Prim's, Kruskal's (+DSU) **This semester, "Prim's" is excluded from our syllabus.**	Dijkstra detailed: https://youtu.be/r9rUUhP2WdM?t=2250 Dijkstra short: https://youtu.be/81TR8arj6ac Prim's and Kruskal's detailed: https://youtu.be/VXv0v7DfROc (There is a small correction in the Kruskal's code written in the description under the video.) Prim's short: https://youtu.be/P1Jj8k8MKrw Kruskal's short: https://youtu.be/-Xkdq8g0UOI	Dijkstra: https://docs.google.com/presentation/d/1Gr2FB43xsF-CFumhG8kOOi6qVvPyUZXNtC4S70iMI_Q/edit#slide=id.g251b994a15a_0_125 Bellman Ford: https://docs.google.com/presentation/d/1YtmzB7aaq-Ufuq7mPna8skVEsWwI53GFYTV_HsOggak/edit#slide=id.g251b994a15a_0_125 Prim's and Kruskal's: https://docs.google.com/presentation/d/1n2SrYJCjcSHWYMa8WUeIASej0LV0GnuPLgYqzgTlUWw/edit#slide=id.g251b994a15a_0_125	MST+Dijkstra: https://drive.google.com/file/d/1zNpTudRm--5wPHKrownXyosZQ1ta69F3/view?usp=sharing
week 9	1. Introduction to greedy, time scheduling interval 2. Fractional knapsack, 3. Huffman encoding decoding **This semester, "Time schedulling interval" is excluded from our syllabus.**	Huffman encoding decoding: https://youtu.be/oAzU9aAwmBA?feature=shared Fractional Knapsack: Short: https://youtu.be/ezNB8Qjf6U4?feature=shared Detailed: https://youtu.be/5xsUlgoefXI	Huffman encoding decoding: https://docs.google.com/presentation/d/1IahoOpRaCw4TDzF58kG0SdiMkuPl1Rq8X9KjMZXjMGo/edit#slide=id.g251b994a15a_0_125 Fractional Knapsack: https://docs.google.com/presentation/d/1-6snzQ-Go_BPvHg_qBdg-XLEqAUYySc6T8zaEg5dU80/edit#slide=id.g251b994a15a_0_125	Greedy Basics, time scheduling: https://drive.google.com/file/d/1U-LLjWQHP-RdARYV8vOYlMkKglZBFzd4/view?usp=sharing Huffman: https://drive.google.com/file/d/1-_oOWBKQARdaWgZvLTUFIV5tCYd5CqZ4/view?usp=sharing fractional knapsack: https://drive.google.com/file/d/1U-LLjWQHP-RdARYV8vOYlMkKglZBFzd4/view?usp=sharing
week 10	1. DP Basics, Knapsack 0/1 2. LCS 3. Coin Change (how many ways, minimum no. of coins) 4. Recursive and iterative DP formulation, comparison **This semester, "Coin change" is excluded from our syllabus.**	0/1 Knapsack: https://youtu.be/5xsUlgoefXI LCS: https://youtu.be/3aCSK5I5iC8?feature=shared (There is a small correction in the Recurrence Relation written in the description under the video.)	0/1 Knapsack: https://docs.google.com/presentation/d/1KadNHy43OfpQt4muFGvG2qO-8daDPpKejzd9zog2Aks/edit#slide=id.g251b994a15a_0_125 LCS: https://docs.google.com/presentation/d/1tWk_hGTWN0UnicWzQNWwPK6zsxC3GIvIdW6pHNAeE90/edit#slide=id.g251b994a15a_0_125	LCS: https://drive.google.com/file/d/1YJonhR6glZoGMLk0G6323MJytohIlvn6/view?usp=sharing DP Basics, knapsack: https://drive.google.com/file/d/1tdh2T7G4L4uMVuLnxvqHhJqVxku8nnoN/view?usp=sharing
week 11	P vs NP **(Self study)**	https://drive.google.com/file/d/1H-hcL4cDZMKq8Ds1JM-ipJD7adjNcmd6/view?usp=share_link Lecture video: https://bracuacbd-my.sharepoint.com/:f:/g/personal/fakhruddin_gazzali_bracu_ac_bd/EkuceTu_illFufQaAcqawGcBfgfdzpODwUFjgGUcBQxnZA?e=0wmLdo	https://drive.google.com/file/d/1H-hcL4cDZMKq8Ds1JM-ipJD7adjNcmd6/view?usp=share_link Lecture video: https://bracuacbd-my.sharepoint.com/:f:/g/personal/fakhruddin_gazzali_bracu_ac_bd/EkuceTu_illFufQaAcqawGcBfgfdzpODwUFjgGUcBQxnZA?e=0wmLdo	https://drive.google.com/file/d/1H-hcL4cDZMKq8Ds1JM-ipJD7adjNcmd6/view?usp=share_link Lecture video: https://bracuacbd-my.sharepoint.com/:f:/g/personal/fakhruddin_gazzali_bracu_ac_bd/EkuceTu_illFufQaAcqawGcBfgfdzpODwUFjgGUcBQxnZA?e=0wmLdo

ZTB Youtube Playlist: https://youtube.com/playlist?list=PL55iLWsFacBEIYQcPH_7smxesY8-xOzMt&feature=shared